Bhavesh Garg

bh05avesh@gmail.com | Website | +44-7405111044 | linkedin.com/in/bhaveshgarg21

EDUCATION

University of Cambridge — Cambridge, UK
MPhil in Machine Learning — *Oct. 2026 - Aug. 2027*

Indian Institute of Technology Bombay — Mumbai, India
Dual Degree (B.Tech., M.Tech.) in Electrical Engineering, Cum. GPA: 8.66 / 10 — *July'17 - July'22*

PUBLICATIONS

1. *Deep Generative Model Based Rate-Distortion For Image Downscaling Assessment* [Paper]
   Yuanbang Liang, **Bhavesh Garg**, Paul Rosin, Yipeng Qin
   Oral presentation *(0.78%)* at Conference on Computer Vision and Pattern Recognition (CVPR 2024)
2. *Perceptual CGAN for MRI Super-Resolution* [Paper]
   Sahar Almahfouz Nasser, Saqib Shamsi, Valay Bundele, **Bhavesh Garg**, Amit Sethi
   Paper accepted at IEEE Engineering in Medicine and Biology Conference (EMBC 2022)

WORK EXPERIENCE

Adaptive Learning for Camera Traps — Remote
Collaborator: Dr. Lukas Picek, Beery Lab MIT — *2026-present*

- Developing an Adaptive Reference Gallery framework using vision foundation models for training-free domain adaptation of wildlife camera trap classifiers

QureAI | AI Scientist — Bangalore, India
Lung Cancer Detection from CT Scans — *March '25 – Feb '26*

- Integrated Test-time Augmentation methods in pulmonary nodule detection pipeline, boosting recall by **14%**
- Led development of CT scan orientation classifier using ResNet model, achieving **97% accuracy** over 3-axis alignment and reduced manual processing

Robot Learning Lab | Research Associate — Freiburg, Germany
Perception for self driving — *June '24 – October '24*

- Leveraged 2D optical flow and 3D scene flow to enhance dynamic scene understanding for Bird's Eye View (BEV) semantic segmentation; utilized RAFT for dense flow estimation

FiveAI | Research Engineer (intern) — Cambridge, UK
Perception for self-driving — *August '22 – Dec '22*

- Investigated the role of **depth estimation** for LIDAR-camera based 3D object detection in self driving systems
- Improved the performance of the in-house camera-only detection model by 20% in the mAP metric

KEY PROJECTS

Super Resolution with Modular Adapters | [Report] — IIT Bombay, India
Master's Thesis, Advisor: Prof. Amit Sethi — *July '21 – June '22*

- Proposed **ModSR**, a modular CNN framework that integrates large-scale pretraining, meta-learning, and lightweight residual adapters to achieve fast and adaptive super-resolution under real-world settings

Cardiff University | Image Downscaling Assessment — *June '21 – Nov. '21*

- Formulated a rate–distortion based image downscaling metric that employs generative SR models to stochastically super-resolve images and measure information preservation in downscaling

ACADEMIC ACHIEVEMENTS

- Recipient of the prestigious **NTSE** and **KVPY** scholarships by the Government of India *('15)*
- Secured All India Rank of **104** in JEE Mains, **389** in JEE Advanced exam out of 1.2 million candidates *('17)*

TECHNICAL SKILLS

Languages/Libraries Python, Pytorch, Keras, OpenCV, Numpy, Pandas, Scikit-learn, C, C++
Software Tools Git, VS Code, PyCharm, MATLAB